UFMF8V-15-3 Space Engineering

Group 4

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Task 2.1 – Orbital Mechanics

Introduction

Within scope of the task 2.1, the estimations have made for the area of interest, swath, payload, ground station, and GPS availability for both our satellite and the ground station. In addition, the satellite's ground track was determined by trading orbital parameters such as altitude, inclination, and eccentricity. This was done to get the desired ground track. Altitude, on the other hand, is the most essential orbital parameter since it influences all aspects of satellites, including their coverage, performance, ground communications. As a result, iteration process has conducted for this research to determine the optimal time to revisit the ground station and the area of interest.

Region of interest

Top 10 Countries that Release the Most Plastic into the Ocean (tons 2021):

Country	MPW released into oceans 2021
Philippines	356,371
India	126,513
Malaysia	73,098
China	70,707
Indonesia	56,333
Brazil	37,799
Vietnam	28,221
Bangladesh	24,640
Thailand	22,806
Nigeria	18,640

Figure 1- Top 10 countries that release plastic to the ocean (World Population Review, 2021).

Both Manila Bay and Puerto Princesa in the Philippines are primary areas that have considered to observe the plastic pollution (Ramos, 2023). Figure 1 illustrates that the Philippines is the country that disposes the most quantity of plastic into the ocean.

Furthermore, in the Philippines, Manila Bay and Puerto Princessa are being investigated as the regions that are identified as having the highest levels of plastic pollution (Cabico, 2024). The majority of the plastics that gathered in Manila Bay were plastics such as plastic bags and plastic bottles (Cabico, 2024). Plastics are the most common materials that are discharged into the manila bay, and they account for around 90 percent of the trash that is collected from that area (Cabico, 2024). in addition, Plastic wrappers, plastic bottles, and nylon fishing lines have contributed to the pollution of the coast side of Puerto Princessa (Cabico, 2024). Therefore, this region of interest is aligned with our mission objective of observing the plastic in Philippine Ocean.

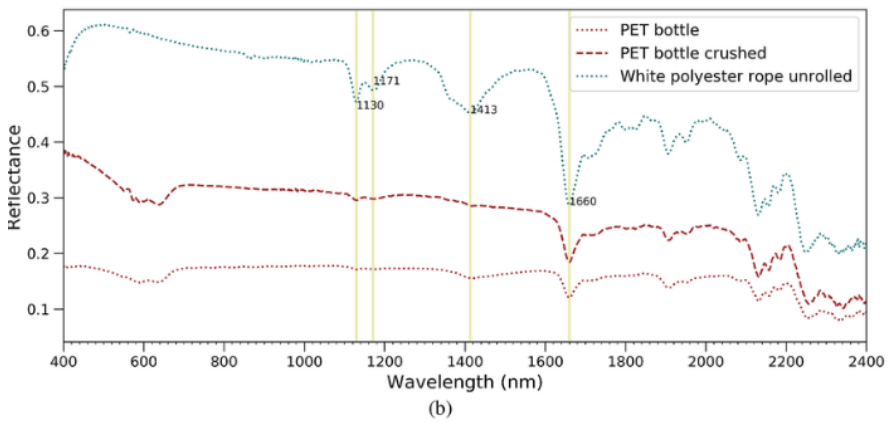


Latitude	Longitude
8.4 deg	116.986 deg
13.934 deg	120.069 deg
14.162 deg	119.68 deg
8.694 deg	116.646 deg

Figure 2-Region of interest and location.

Payload

Spectrometers were chosen initially as the payload of the mission to detect plastics. However, the complexity of the spectrometer resulted in changing the spectrometer to a camera for this mission. The wavelength of the plastic was investigated in the study to find the camera. Therefore, the wavelength was estimated that 650 nm as shown in figure. Then the iSIM-90 VNIR SWIR Camera was taken as the payload in the satellite after a thorough study and the main reason to choose this was the significant specs as illustrates in the following table. In addition, this camera can detect plastic with its VNIR and SWIR band which can detect wavelength of 650 nm.



(b)

Spectral reflectance of dry virgin (a) unrolled Polypropylene (PP) ropes, a Low-density polyethylene (PE-LD) cup (b) Polyethylene terephthalate (PET) water bottles and a Polyester (PEST) rope. Absorption features are highlighted by the vertical yellow lines.

Figure 3 (Moshtaghi et al., 2021)

iSIM-90 VNIR SWIR Camera Specs	Value
Size	115*218*308 mm
Mass	<6kg
Band	PAN, RGB, VNIR, SWIR
Swath	13 km @ 500km, 26 km @ 500 km
Focal length	775 mm

Table 1 -(Sat search, 2023)



Figure 3- isim-90 camera (Sat search, 2023)

After examining the camera specs it's important to calculate the pixel size of the camera to determine the resolution of image. Moreover, following table illustrates important parameters in the payload.

Mission parameters	value
Swath	40 km
GSD	1.5
Angular resolution	1.0313 arcsec
Diameter	0.15 cm
Pixel size	3.875e-5

Table 2

GSD and Swath parameters play a crucial role in a space mission to achieve its goals. Therefore, the swath width was taken as 40 km because to specifically target that coast area and to not interrupt with the other countries that polluting in the ocean. A 1.5m GSD was selected as this would fit the largest types of plastics found in the ocean making it easy to identify in an individual pixel (odonhoue, 2023).

Following MATLAB code was used to calculate some of the parameters in the above-mentioned Table.

payload

```
H=300e3;% altitude
GSD=1.5;
AR=atan(GSD/H);%angular resolution in rad
ARdeg=rad2deg(AR);% angular resolution in deg
ARarc=206265*(GSD/H);
lamda=650e-9;%wavelength
D=251643*(lamda/ARarc); %camera diameter
Focallegth= 0.775;%m
pixelsize=(GSD*Focallegth)/H;
```

Figure 5 - payload element MATLAB code.

Orbital trade-off

Orbit trade-off is crucial in this study to get the best revisit time on the region of interest and ground station location. As mentioned in the introduction Altitude has considered as the main orbital element among other orbital elements such as eccentricity and inclinations. Therefore, iterations have conducted of revisits and total time durations align with the ground station and region of interest.

Altitude Vs Coverage

Altitude(km)	No of visits to ground station	Total time duration at ground station (s)	No of visits to region of interest	Total time duration at ROI (s)
300	6	2968.731	4	1800.058
400	5	3077.134	4	2102.775
500	6	3957.452	5	3130.491

Table 3 -Alt vs Coverage results from STK.

From the Altitude study, with each increase of altitude the coverage is getting improved but the as for the mission the most suitable altitude is 300 since the GSD value is smaller. In addition, with increase of altitude the GSD value also increases. Therefore, the resolution of the image will reduce.

Eccentricity Vs Coverage

Eccentricity	No of visits to ground station	Total time duration at ground station	No of visits to region of interest	Total time duration at ROI
0.3	5	4281.625 s	5	7350.143 s
0.5	4	5420.442 s	6	11217.40 s

Table 4 - Eccentricity Vs Coverage from STK.

The altitude iteration was done by the eccentricity of 0 and then eccentricity was iterated with increasing the eccentricity value. Therefore, the coverage has increased.

Inclination VS coverage

Inclination	No of visits to ground station	Total time duration at ground station	No of visits to region of interest	Total time duration at ROI
49º	6	4004.099 s	4	7494.868 s
47º	5	3770.387 s	4	7573.559 s
52.5º	5	4281.625 s	5	7350.143 s

Table 5 - Inclination VS coverage from STK

Using the 0.3 eccentricity and 300 km as the altitude in this stage we have iterated the inclination angle and then found the best inclination angle for our mission.

The other orbital parameters such as argument of perigee, RAAN, True Anomaly were taken as a fixed value due to slim geography of the Philippine. In addition, the following parameters gave best coverage whilst considering our thin ROI.

Orbital Element	Value
Semi major axis	6678.14km
Eccentricity	0.3
Inclination	49º
Argument of perigee	0
RAAN	-4º
True Anomaly	200º

Table 6 - Orbit elements for the study

Field of view change with the altitude

From the Field of view study, it's evident that with each increase in altitude field of view angle changes because the FOV angle linked with the swath and altitude. When the altitude increases the FOV angle decreases.

Altitude (km)	Swath(km)	FOV °
300	40	7.59
400	40	5.71
500	40	4.57

Table 7- change of FOV with altitude

Location of ground station

The ground station that has selected (Cobbett Hill Earth Station), as well as the other ground stations, were selected in a strategic manner, taking into consideration elements such as the orbital route, and the coverage requirements. According to the orbital parameters presented in the section on orbital trade off, Cobbett Hill Earth Station had the highest number of visits and the longest period of stay. An initial selection was made of three ground stations, and the ground station with the greatest coverage was selected.

Ground station	No of visits	Total time duration
Cobbett Hill Earth Station	6	4004.099
Malargüe- station Argentina	5	**3781.09**
New Norcia - station Australia	5	3416.032

Table 8 - ground stations vs coverage

GPS availability

In this task the GPS availability was examined using STK software for both satellite and ground station. The major aim in here is to monitor the location of the ground station with GPS and the satellite. In addition, coverage access has done to see the GPS availability on both space craft and ground station. Therefore, the following figure is shown the access with the satellite and ground station.

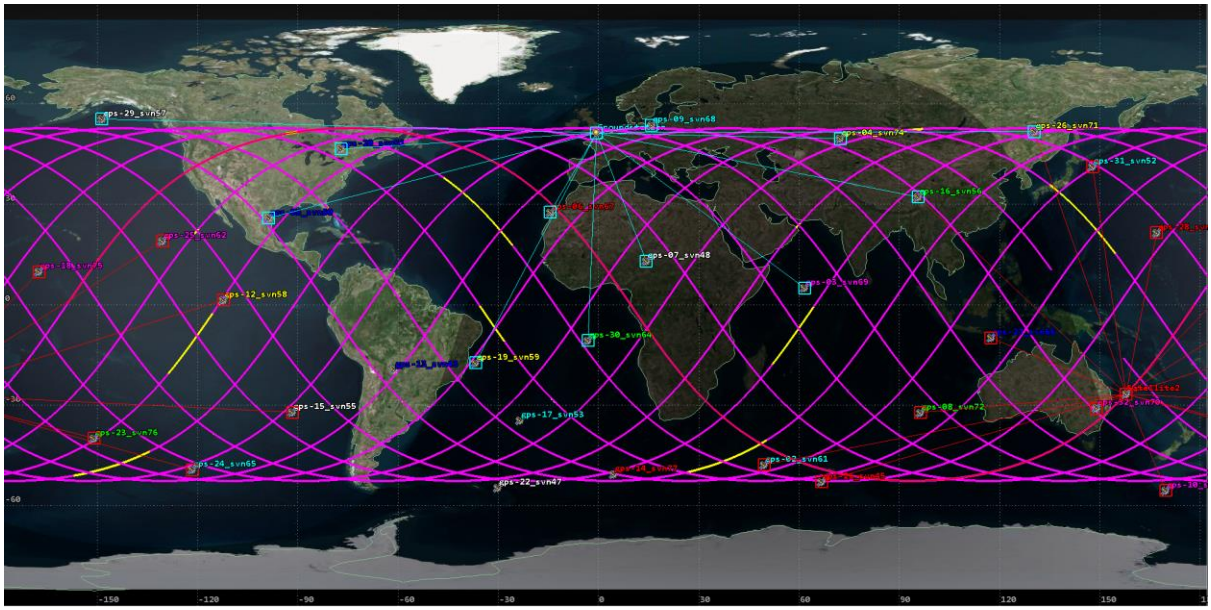


Figure 6 - GPS availability for Satellite and Ground station.

2.2 Platform Stability

Aims and Objectives

- To observe with a payload instrument with reasonable precision to make good use of the measurement.
- To explore the effects of disturbances within a dynamic environment throughout the orbit propagation.
- Size the actuators dependent on the spacecrafts size and mission profile whilst in a closed loop compilation of STK and MATLAB/Simulink

Attitude and Orbit Control Subsystem (AOCS)

Platform stability has a crucial role in determining the success of a satellite mission. Platform stability refers to the competency of the satellite in maintaining a constant orientation or attitude in orbit.

Attitude knowledge and Attitude Accuracy

The following parameters were used frequently throughout the attitude control system equations:

Pointing Accuracy (rads)	Pointing Knowledge (rads)
5e-6	2.5e-6

Table 9

Pointing knowledge and pointing accuracy will ensure the satellite's on-board instruments, such as the sensors and antennas, observes the ground station, in Cobbett Hill, UK, and is able to capture imaging from our desired area target along the west coast of the Philippines with reasonable accuracy.

Sources of Disturbances

The four main disturbances that were considered in this report that affected torque were: drag, geomagnetic, gravity gradient and solar pressure. The satellite was designed with worst case scenario in mind therefore, a combination of the greatest total torques in the x, y and z were simulated in preparation for the worst-case scenario.

The satellite shape was carefully picked to be a cylinder due to exploit spinning inertia as well in the future having the adaptation to be a dual spin satellite [Ovchinnikov, 2021]. The height is 2.5m and a 1m radius, big enough to carry the payload. Next, moments of inertia were found using the following formulas: (1)

$$I_z = I_{\parallel} = \frac{mr^2}{2} \qquad I_x = I_y = I_{\perp} = \frac{m}{12}(3r^2 + h^2)$$

$I_z = 50kgm^2$ $I_x = I_y = 58.33kgm^2$

All disturbances consider are in the following table:

Axis	Drag (Nm)	Geomagnetic (Nm)	Gravity (Nm)	Solar Pressure (Nm)	Total (Nm)
x	4.08E-4	4.67E-5	2.92E-6	0	4.57E-4
y	8.16E-4	4.67E-5	2.92E-6	0	8.66E-4
z	1.4E-3	4.67E-5	2.92E-6	1.93E-4	1.7E-3

Table 10

In the event of a worst-case scenario, the satellite is ought to turn 180° is 60 seconds, giving a Slew rate $\dot{\theta}$ of $\frac{0.0524\text{rads}}{sec^2}$. Thereafter a radius was chosen for the wheel sizing of $60cm$ and this produce a mass for the x and y wheel of 22.9g and mass in the z wheel of 19.6g this was added to the Simulink closedloopfunction.

An Attitude controller was needed to control the orientation of the spacecraft as these disturbances can cause the satellite to steer off course. As well as the attitude controller is used for better coverage over the ground track and acquiring and downlinking data from the earth station as this allows the satellite to move in all 3 axis.

To do this celestial sensing is used the determine the position relative to bodies such as the sun. The sun is the primary source of attitude determination as there are light sensors which produce less resistance in illuminated light levels. However, this can only be measured in two axis at any time.

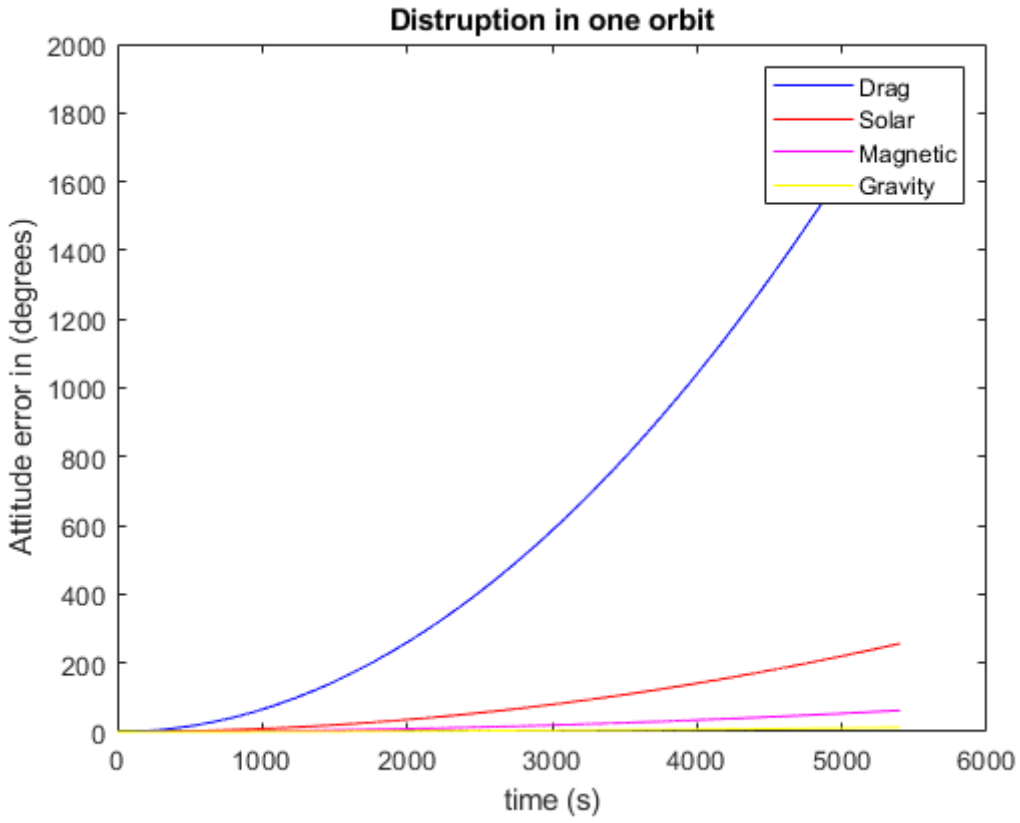


Figure 7

From this graph, disturbances can lead to attitude error therefore it is important to always account for this within the applied torque to counteract this angular momentum. Here the Drag value seem to be a factor of 80 out of the expected typical attitude error.

Results and discussion

The simulation was run, and the ROI (Region of Interest) was used to demonstrate the attitude controller. Figure # shows the angular speed of the spacecraft, initially the facing the Nadir then transitioning to look at the region of interest. The satellite begins to move when the ROI is 63.8° and 700km away. The Satellite will then take 30 seconds to accelerate to a

constant speed as well decelerate, the first peak demonstrates this. Whilst looking at the target, after passing the midpoint, the spacecraft is this rotated 180° to take images of the second half of the ROI, this is shown with the largest peak on the graph. Lastly the final peak shows moving away from the RIO back to the Nadir, following this, the small amount of attenuation with some dampening. However, the rate of dampening in not constantly decreasing implying an error in the PID controller.

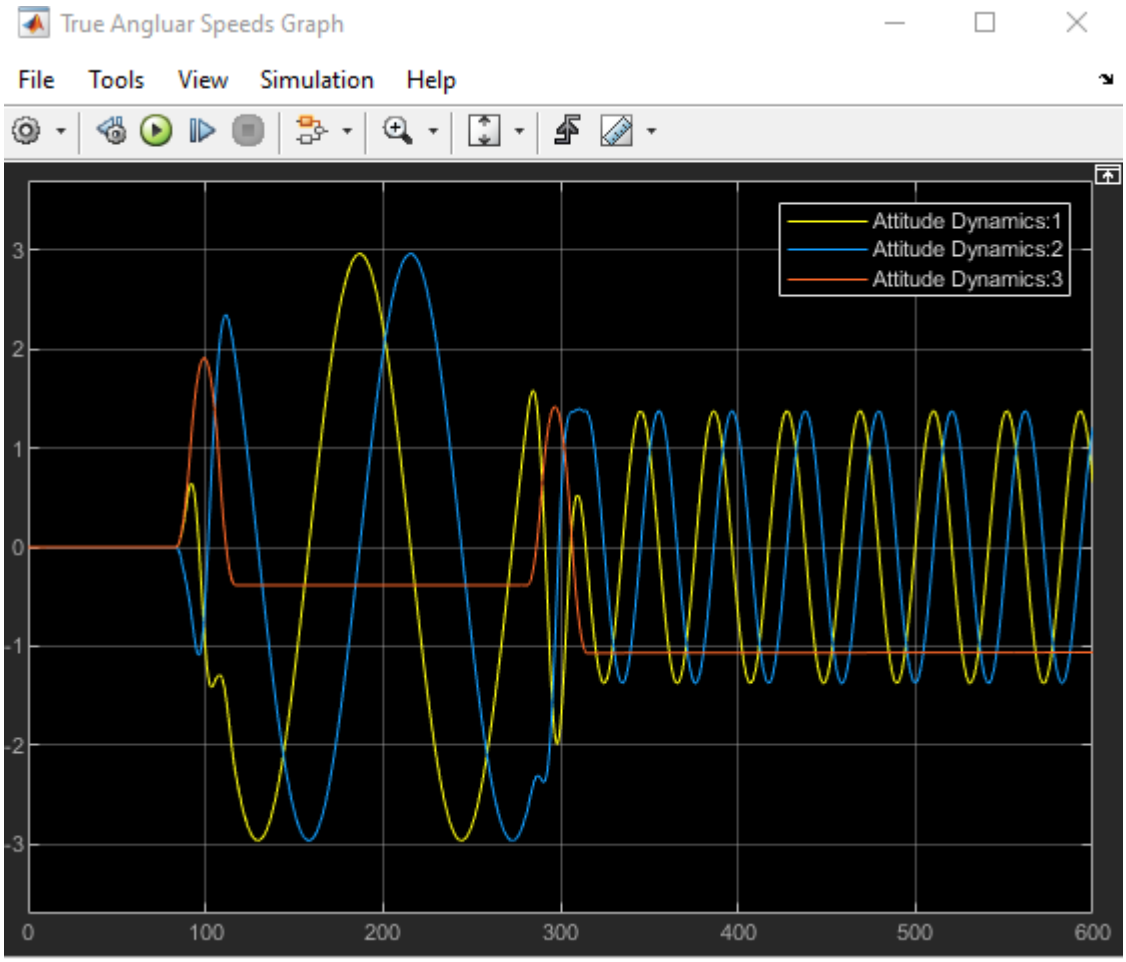


Figure 8

Figure 9 and Figure 10 demonstrates the difference between reference and true before the PID controller. The reference simulates a bang-bang controller where the pulses are ideal and instant, whereas in real life there will be ramp for where there is angular acceleration and deceleration.

Figure 9

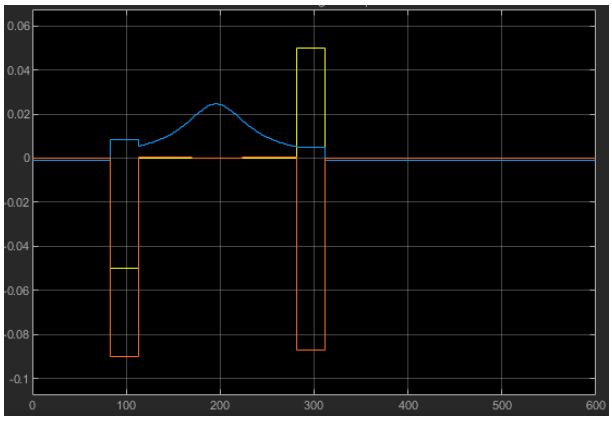
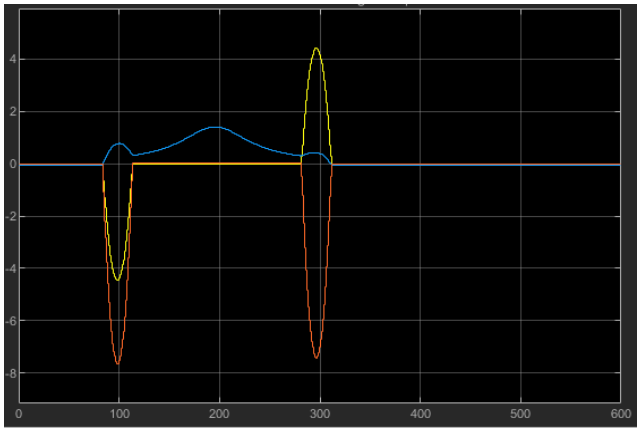


Figure 10



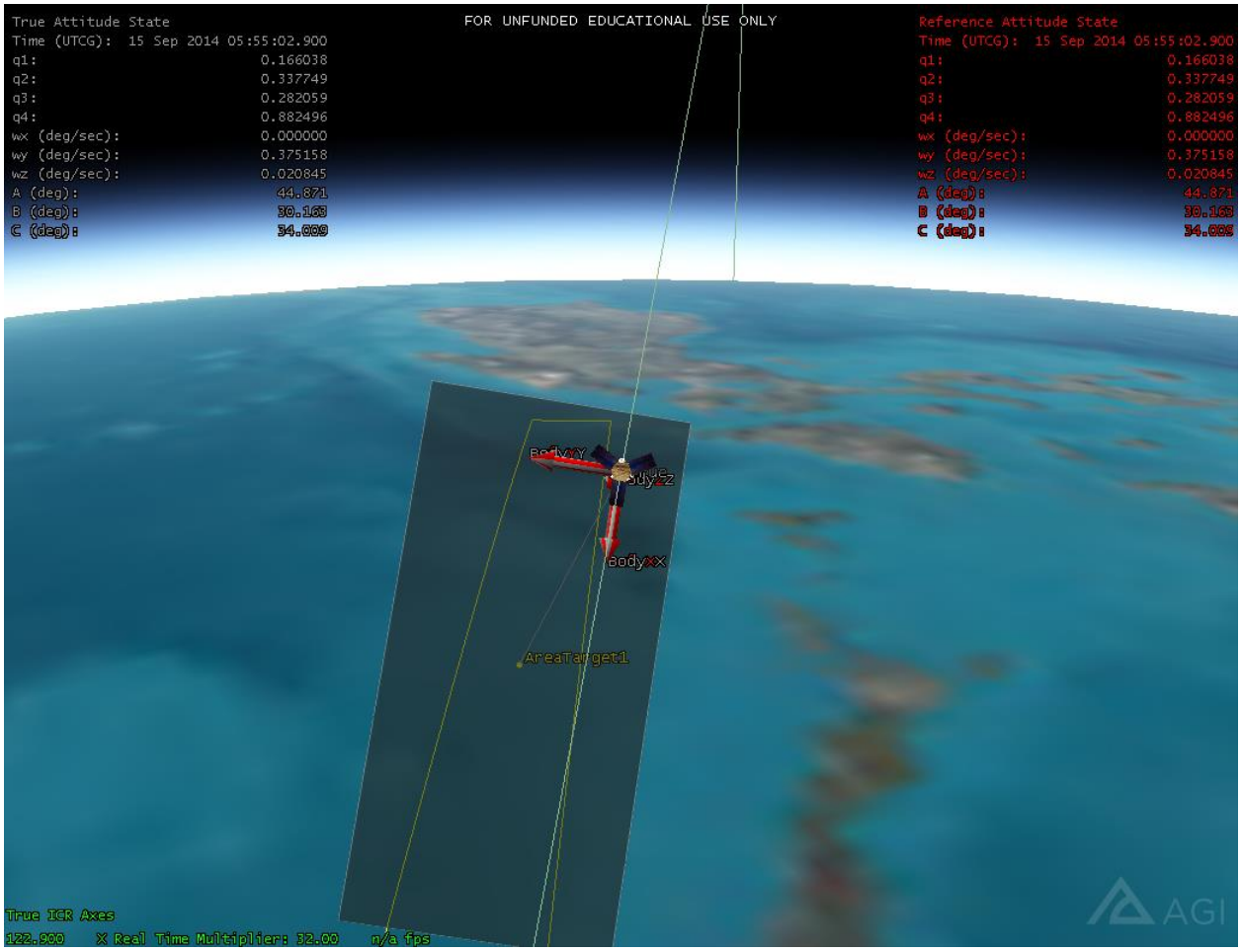


Figure 11

Figure 11 is showing the very start of the observation with the start of the viewing angle being 44°,30°, 34° in the x,y and z direction respectively from the z axis facing the nadir moments before.

In Conclusion observer payloads were used with reasonable measurement for the purpose of litter waste in the ocean so prevent damage to aquatic ecosystems. This Report aims to find a trustworthy source to find the most polluting oceanside towns and villages. The effects of disruption were explored in a dynamic simulation to, and the actuators were appropriate size in a disaster scenario. This section demonstrates, using stimulatory software such as STK and MATLAB/Simulink, how we achieved this platform stability for our satellite. Future work would be conducted on detumbling in fast speeds.

2.3 communication

Introduction

Communications is the most important aspect of in any space mission; without it completing achieving the mission objectives would be impossible. There are many design decisions that must be carefully selected such as determining the data size of the science acquisitions for setting an appropriate the upload and download rate. Also considering which frequency band is required for the data science requirements.

The communication system will include two download types. Telemetry, which uses low frequencies to downlinks the voltage, current, temperature and sensor data to the ground station, whilst the telecommand uplink uploads commands or mission tasks to the satellite. However, for payload or science data that are required to be downloaded it will require a higher data rate channel to handle larger quantities of information.

A process that integrates link budget is initiated when commands from the ground station are sent to the satellite. Link budgets are essential calculations that help to ensure the robustness of communication. These calculations determine many key components from the data size of the payload to the required size of the shape both on in space and on the ground.

The report will demonstrate the refinement of the link budget through calculating the static link, where the mission's assets are fixed, then using a simulation tool called STK to analyse the dynamic link due to the change in geometry and finally a detail dynamic link that utilises the satellite link budget link analyser app on MATLAB.

Spectrometer and Data Rate Calculations

Figure 4 is multispectral payload that has two channels allowing imaging in Visible and Near Infrared (VNIR) and Short-Wave Infrared (SWIR) simultaneously allowing for a variety of plastic to be identified. The payload originally comes with data storage of 500 GB however due to our target area being so large, (Area = 2.8E+10 Km^2) it was chosen to request for an upgrade as according to 'iSIM-90 | satsearch' (n.d.). These two channels provide an enough capacity to capture imagining of plastic pollutants in the ocean as shown in figure 3. This shows that the general trend is that most reflectance around 400nm-600nm which fits comfortably in the VNIR (Moshtaghi et al., 2021).

The ISIM-90 spectrometer's specifications were obtained, and the data rate per second was computed to ascertain the transmission efficiency of the data container. It is necessary to first calculate the orbital and ground velocities. The orbital velocity was 9.55E-1 m/s, and the ground velocity was 9.12E-1 m/s. The spectrometer's eight bits and eight bands were also mentioned in the specification. Based on these, the data rate for the VNIR GSDs observed by the swath was calculated to be 9.90E-1Mbps, which is both sustainable and slower than anticipated; yet the data rate would also increase if the bits were raised. The second channel's (SWIR) data rate was then calculated through the VNIR data rate producing a speed of 2.47E-1 Mbps, which is less than expected too. As this is a spectral range is only an optional addition there is no need for SWIR so using two VNIR Spectrometers instead. This will give the spectrometer an aggregate data rate of 1.98E+0 Mbps.

Link Budget – Initial Calculations

Link Budget			
Boltzmann Constant	k	1.31E-23	
Reciever Temperature	T	1.69E+01	deg Celcius
Signal Bandwidth	B	7.31E+04	Hz
Satellite Transmitting Power	P_T	5.00E+00	Watts
Satellite Transmitting Gain	G_T	1.00E+01	Watts
Ground Station Receiving Gain	G_R	5.00E+00	Watts
Bit Rate	R_b	4.03E-03	bps
Free Space Path Loss	FSPL	1.48E+01	
Effective Isotropic Radiated Power	EIRP	2.00E-01	Watts
Signal Power	S	3.96E-13	Watts
Noise Power	N	1.61E-17	
SNR		2.46E+04	dB

Energy per Bit	E_b	9.82E-11	W/bits
Noise Density	N_o	2.20E-22	W/Hz
energy per bit relative to noise power spectral density	E_b / N_o	4.46E+11	dB
Carrier to Noise Ratio	C/N	1.89E+01	dB
Bit Error Rate	BER	0.0E+00	dB

Table 11 - Link Budget

The link budget demonstrates its quality of the link through figures of merits as shown in table 11 (in Bold). The four merits are the Signal to Noise Ratio (SNR), the Carrier to Noise Ratio (C/N), the energy per bit relative to noise power spectral density (E_b/N_O) and finally the Bit Error Rate (BER).

The SNR was calculated through the finding S, power reciever antenna:

$$S = \frac{P_T G_T G_R \lambda^2}{(4\pi R)^2} \ (W)$$

Then finding N, noise power:

$$N = kTB$$

Then dividing it by one another to calculate SNR:

$$SNR = \frac{S}{N}$$

The Value produced in table ? produced a high ratio suggesting the signal is strong even through noise disturbance.

The E_b/N_O is the signal to noise power ratio per bit producing a finer and more accurate performance metric, thus allowing to determine the BER in the link. This is calculated through the equation:

$$E_b = \frac{S}{R_b}$$

$$N_O = kT$$

$$\frac{E_b}{N_O} = \frac{S}{R_b kT}$$

This also produced a high ratio as shown in table 11.

SNR is a signal after modulation when the carrier signal and the modulating signal are mixed. However, before modulation, the carrier-to-noise ratio is measured.

$$\frac{C}{N}(dB) = 10\log\left(\frac{E_B}{N_O}\right) - 10\log\left(\frac{f_b = R_B}{B}\right)$$

This equation produced a ratio that is not as high as the other two ratios in table ? however, this doesn't mean that the carrier signal is not suitable.

$$BER = \frac{1}{2}erfc\left(\sqrt{\frac{E_b}{N_O}}\right)$$

As all the ratios are high we expect the BER to be very low. Aligning with Proakis's (2001) use equation #, the Bit Error Rate was 0.00E-0 which is measured to little to no transmission errors.

Link Budget – STK

Link budget must ensure the communication process of satellite to ground station or ground station to satellite works efficiently.

Carrier to noise ratio is a crucial parameter in the link budget which address the ratio of carrier signal power to noise power density. And in the following link budget it was found to be for an effective scenario as a 35 dbhz. That means it has a good GPS reception.

Bit error rate is also a key parameter to monitor the quality link of a communication system and as general rule a BER at least 10e-5 is desirable. In the space mission.

These two are the important parameters that considered in this STK link budget, and variables used such as antenna gain, transmit power, antenna diameter, data rate, frequency to achieve the link budget.

The following BER rate graph was achieved from the analysing the link budget on STK by changing the receiver parameter and transmitter parameters and important thing noticed in the process was that when the transmit power decreases carrier to noise ratio also decreases and when the diameter decreases the carrier to noise ratio and BER decreases.

GPS Jamming

Radio communications are interefered using a tansmitting object and that process is addressed as the GPS jamming.

Volume Graphics

GPS jamming is a key parameter in identifying the satlites that not working with access to ground station. In addition, placing a transmitter near to the ground station will interfere with ground stations radio frequency.

References

‘iSIM-90 | satsearch’ (no date) satsearch.co [online]. Available from: https://satsearch.co/products/satlantis-isim-90. [accessed

Moshtaghi, M., Knaeps, E., Sterckx, S., Garaba, S. and Meire, D. (2021) Spectral reflectance of marine macroplastics in the VNIR and SWIR measured in a controlled environment. Scientific Reports [online]. 11 (1).

Ovchinnikov, M. (2021) Magnetic attitude control and periodic motion for the in-orbit rotation of a dual-spin satellite. Available at: https://www.sciencedirect.com/science/article/pii/S0094576521002721 (Accessed: 07 May 2024).

Proakis, J.G. (2001) Digital communications. Boston, Mcgraw-Hill.

Cabico, G.K. (2024). *Plastic makes up 90% of Manila Bay litter — monitoring survey*. [online] Philstar.com. Available at: https://www.philstar.com/headlines/climate-and-environment/2024/01/26/2328694/plastic-makes-90-manila-bay-litter-monitoring-survey [Accessed 9 May 2024].

Ramos, D. (2023). *Plastic Pollution in the Philippines: Causes and Solutions*. [online] Earth.org. Available at: https://earth.org/philippines-plastic/ [Accessed 9 May 2024].

Sat search (2023). *iSIM-90 | satsearch*. [online] satsearch.co. Available at: https://satsearch.co/products/satlantis-isim-90 [Accessed 9 May 2024].

World Population Review (2021). *Plastic Pollution by Country 2020*. [online] World Population Review. Available at: https://worldpopulationreview.com/country-rankings/plastic-pollution-by-country [Accessed 9 May 2024].

odonhoue, D. (2023). *Understanding Ground Sampling Distance (GSD) - May 9, 2024*. [online] Map Scaping. Available at: https://mapscaping.com/understanding-ground-sampling-distance-gsd/ [Accessed 9 May 2024].